Description

The NASA Commercial Modular Aero-Propulsion System Simulation (C-MAPSS) dataset comprises simulated run-to-failure data for commercial turbofan engines. It is extensively utilized in prognostics and health management (PHM) research, particularly for developing and evaluating models that predict the Remaining Useful Life (RUL) of engines.

Figure 1 provides a schematic of the engine simulated using C-MAPSS, which models an engine with a thrust rating of 90,000 lb. Experiments were conducted under conditions ranging from sea level to 42,000 ft in altitude, Mach numbers from 0 to 0.84, and throttle angles from 20 to 100 degrees. Users can adjust three conditions: flight altitude, Mach number, and throttle angle, to simulate different environmental conditions. The model generates 21 output signals that can be analyzed.

The dataset includes four subsets—FD001, FD002, FD003, and FD004—each representing different combinations of operating conditions and fault modes. Each subset contains multivariate time-series data collected from multiple engines, with each engine operating until failure. The data includes sensor measurements and operational settings recorded at each time cycle. The training sets provide complete run-to-failure trajectories, while the test sets contain partial trajectories, and the corresponding RUL values are provided separately.

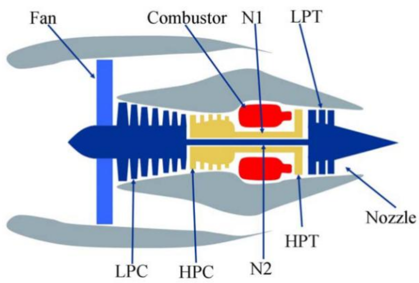

Figure 1. C-MAPSS simplified engine simulation model

Size

Approximately 43 MB

Platform

The dataset is provided in plain text (.txt) format and is platform-independent.

Environment

Any operating system with support for text file processing.

Major Component Description

The data are provided as a zip-compressed text file with 26 columns of numbers, separated by spaces. Each subset (FD001–FD004) includes:

- **Training Data** (train_FD00x.txt): As shown in TABLE 1 and TABLE 2, each row is a snapshot of data taken during a single operational cycle, each column is a different variable, containing:
  1）Engine ID;
  2）Cycle number;
  3）3 operational settings;
  4）21 sensor measurements,
- **Testing Data** (test_FD00x.txt): Similar structure to the training data but for partial sequences ending before failure.
- **RUL Labels** (RUL_FD00x.txt): Provides the true remaining useful life for each engine unit in the test set.

TABLE 1. Column Parameter Names of C-MAPSS datasets

Column Number	1	2	3-5	6-26
Parameter Name	Engine ID	Engine Cycle Number	Operating Conditions	Sensor Data

TABLE 2. Monitoring parameters of different sensors output by C-MAPSS

Number	Symbol	Description	Units
S1	T2	Total temperature at fan inlet	°R
S2	T24	Total temperature at LPC outlet	°R
S3	T30	Total temperature at HPC outlet	°R
S4	T50	Total temperature at LPT outlet	°R
S5	P2	Pressure at fan inlet	psia
S6	P15	Total pressure in bypass-duct	psia
S7	P30	Total pressure at HPC outlet	psia
S8	NF	Physical fan speed	rpm
S9	NC	Physical core speed	rpm
S10	EPR	Engine pressure ratio (P50/P2)	--
S11	PS30	Static pressure at HPC outlet	psia
S12	PHI	Ratio of fuel flow to Ps30	pps/psi
S13	NRF	Corrected fan speed	rpm
S14	NRC	Corrected core speed	rpm
S15	BPR	Bypass Ratio	--
S16	FARB	Burner fuel-air ratio	--
S17	HT_BLEED	Bleed Enthalpy	--
S18	NF_DMD	Demanded fan speed	rpm
S19	PCNFR_DMD	Demanded corrected fan speed	rpm
S20	W31	HPT coolant bleed	lbm/s
S21	W32	LPT coolant bleed	lbm/s

The subsets(FD001–FD004) differ in the number of operating conditions and fault modes, as summarized in the TABLE 3:

TABLE 3. Overview of FD001 – FD004: Operating Conditions and Fault Modes

Data set	FD001	FD002	FD003	FD004
Train Trajectories	100	260	100	249
Test Trajectories	100	259	100	248
Operating Conditions	1	6	1	6
Fault Conditions	1	6	2	2

Contact information

If users have questions regarding the data, contact with *yuanyuan.gao@nuaa.edu.cn*

Reference: A. Saxena, K. Goebel, D. Simon, and N. Eklund, "Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation", in the Proceedings of the Ist International Conference on Prognostics and Health Management (PHM08), Denver CO, Oct 2008.